

35801: Corporation Finance

Course Presentation

Pietro Veronesi
The University of Chicago
Booth School of Business

Who am I?

- Chicago Board of Trade Professor of Finance
 - Affiliated with NBER and CEPR
- Education
 - 1997- Ph.D. in Economics - Harvard
 - Dissertation: “Stock Market Volatility in Regime Shifts Models”
 - 1993 - M.Sc. in Econometrics - LSE
 - 1992 - Laurea in Economics - Bocconi
- Research
 - Stocks and bond valuation under uncertainty, option pricing
 - Optimal IPO timing and IPO waves
 - CEO compensation, incentives, and asset prices
 - Uncertainty about government policies and asset prices
- Teaching
 - Investments, Corporate Financial Risk Management, Fixed Income Securities, Financial Instruments, Option Pricing
- Consulting
 - Compass Lexecon, Axegine LLC, Lincoln International

Course Material

- Canvas site: <https://canvas.uchicago.edu/courses/62586> (EXP)
<https://canvas.uchicago.edu/courses/62581> (AXP)
- Course information/mechanics
 - Course schedule/syllabus
 - Announcements
 - Notes/case questions
 - Problem sets/solutions
 - Additional readings
- Text: Berk and DeMarzo, *Corporate Finance*

Course Requirements

- Final grade based on
 - Final Exam
 - Midterm (“optional”)
 - Case Writeups
 - Participation
- Four cases
 - Write up → answer the questions
- Class participation
 - Constructive
 - Not distracting colleagues (e.g., being on time)

Administrative Information

- Groups for cases and problem sets
 - One write-up per group
 - Exact answers to case studies will not be posted or distributed
- You MUST take midterm and final exam in class
- You MUST adhere to the Booth HONOR CODE

Administrative Information

- Office Hours: By Appointment
- Teaching Assistant:
 - Suri Chen (AXP: shiruic0@chicagobooth.edu)
 - Kostantinos Vogklis (EXP: kvogklis@alumni.chicagobooth.edu)
 - Content questions: first post on Ed Discussion or email TA
- TA will conduct review sessions
 - Not mandatory, but recommended
 - Will send “rough” plan before each review session
 - Recommend you look at the review session questions before the review session
 - Will be available on site all day

Other Administrative Information

- Readings:
 - Lecture notes
 - Comprehensive and include parts for self-study
 - Textbook
 - Minimum sections to read in course schedule
 - Additional Readings
 - Newspaper articles, white papers, academic articles
- Problem Sets (including problems from BD):
 - Problems and Solutions will be posted
 - Don't hand in problem sets, but critical to do!
 - Discussed in review sessions

What Are We Hoping to Accomplish?

Two Parts to the Course...

Part I: Capital Budgeting

- What real assets should the firm invest in?
- Project valuation
- PV, discount rates, valuation in practice (WACC and APV)

1

Valuation Basics

- Free Cash Flows, Discount Rates, and NPV

2

Direct Applications

- Cost of Capital
- WACC and APV

3

Asset Pricing/Investment Fundamentals

Two Parts to the Course...

Part II: Financial Policy

- How should projects be financed?
- Debt, equity, information problems, taxes
- Payout policy

1

Financial Policy Basics

- The “five” assumptions of Modigliani-Miller

2

Debt or Equity?

- Debt Tax Shield vs. Wealth Transfer/Debt Overhang/Risk Shifting

3

Dividends or Cash?

- Signaling/Catering/Buffer

The Two Parts Come Together...

Part I: Capital Budgeting

- What real assets should the firm invest in?
- Project valuation
- PV, discount rates, valuation in practice (WACC and APV)

Part II: Financial Policy

- How should projects be financed?
- Debt, equity, information problems, taxes
- Payout policy

Part III: Putting it all together...cases...

- Making capital budgeting and valuation decisions
- Making financial policy decisions

What you should be able to do?

How much is a Firm/ Project worth?

Aramco Valued at \$1.7 Trillion in World's Biggest IPO

Saudi oil giant's IPO breaks record set by Alibaba Group in 2014

By [Ben Dummett](#) and [Summer Said](#)

Updated Dec. 5, 2019 2:41 pm ET

Saudi Aramco priced its initial public offering Thursday at the high end of the targeted range to give the oil giant a total value of \$1.7 trillion in the world's biggest-ever IPO.

EXCLUSIVE STOCKS

Reddit Eyes Valuation of Up to \$6.5 Billion in IPO

The social-media company was valued much higher a few years ago

By [Corrie Driebusch](#) [Follow](#)

Updated March 1, 2024 4:01 pm ET

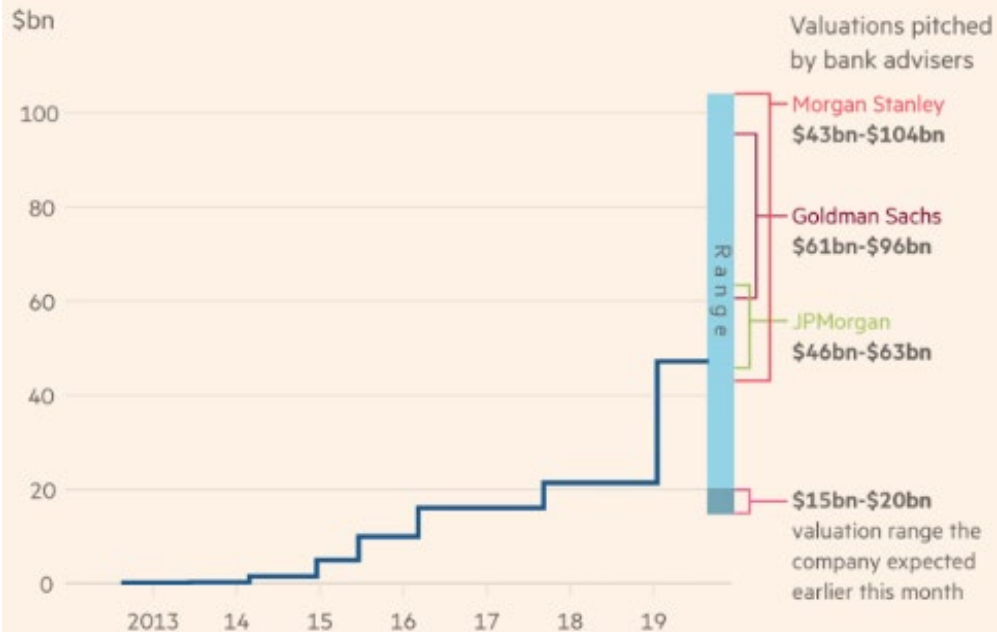
Mar 1, 2024 – Wall Street Journal

WeWork turmoil puts spotlight on JPMorgan Chase and Goldman Sachs

Eric Platt, Andrew Edgecliffe-Johnson, James Fontanella-Khan and Laura Noonan in New York

SEPTEMBER 24 2019

We's wide-ranging valuations



Sources: FT research; Crunchbase

© FT

What you should be able to do?

Issue Debt or Equity?

Why Acadia Pharmaceuticals Is Plummeting

Shares take a dive after management announces a fresh capital raise. Here's what investors need to know.

Brian Feroldi ([TMFTypeoh](#))

Nov 27, 2018 at 3:22PM

What happened

After management announced its intention to raise capital through a common stock offering, shares of **Acadia Pharmaceuticals** ([NASDAQ:ACAD](#)), a commercial-stage biopharma focused on diseases of the central nervous system, fell 12% as of 3:07 p.m. EST on Tuesday.

8,564 views | Mar 30, 2018, 02:52pm

Can Tesla Absorb An Equity Raise Without Cratering The Stock



Chuck Jones Contributor

I cover technology companies, worldwide economies and the stock market

BUSINESS NEWS APRIL 30, 2013 / 2:05 PM / 6 YEARS AGO

Apple wows market with record \$17 billion bond deal

NEW YORK/LONDON, April 30 (IFR) - Apple Inc wowed the debt markets on Tuesday with the largest non-bank bond deal in history, pricing a whopping \$17 billion as the U.S. computer giant switches strategy to placate restless shareholders.

What you should be able to do?

To Pay or Not to Pay?

Not Paying Cash



**Apple now has
\$243.7 billion in
cash on hand**

Sara Salinas | @saracsalinas

Published 4:33 PM ET Tue, 31 July 2018

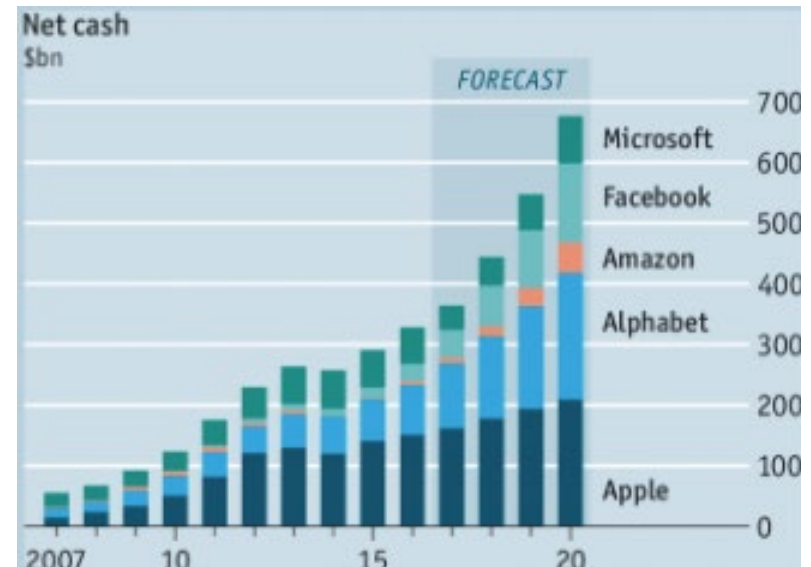


Payout Cash



Pimco hoards cash to avoid risks after Trump rally

Big US fund manager prepares for turbulence as it predicts market euphoria will fade



The Economist, June 3 2017

What you should be able to do?

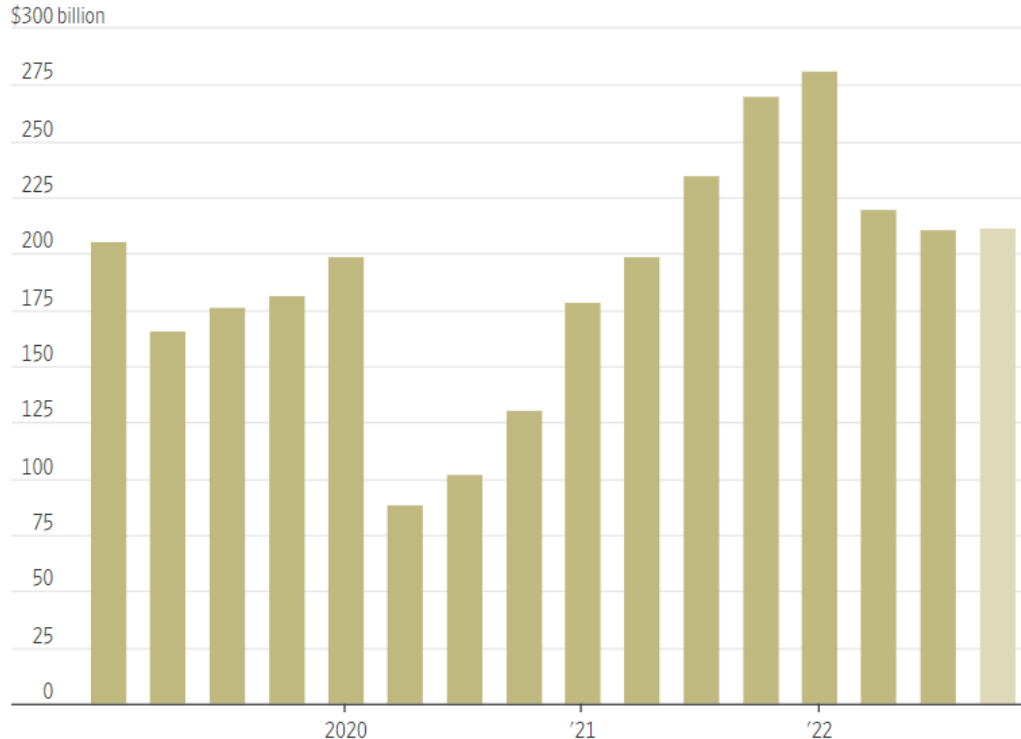
To Repurchase or Pay Dividends?

Mar 21, 2023 at 2:53 pm ET ★

Buybacks from S&P 500 Companies Set Record in 2022

By [Hannah Miao](#)

S&P 500 companies, quarterly buybacks



Source: Wall Street Journal

Stock buybacks exploded after the tax cuts. Now they're slowing down

By [Matt Egan](#), CNN Business

Updated 7:56 AM ET, Thu August 22, 2019

Stock Buybacks: What Every Investor Needs to Know

They have been attacked by many academics and progressive politicians. Now, with a new administration, the battle could soon get even more heated.

By [Randall Smith](#)

Dec. 5, 2020 11:31 am ET

Meta Stock Jumps After Facebook Parent Initiates Dividend



By [Candice Choi](#), Editor

Feb 1 2024 -- Wall Street Journal

Shares of Meta jumped in after-hours trading after the company reported its results and said it's initiating a quarterly dividend of 50 cents per share.

The stock was recently up more than 11% after closing up more than 1% during the regular session. Get more on [Meta's quarterly results here](#). Follow the Journal's coverage of today's [big tech earnings here](#).

What you should be able to do?

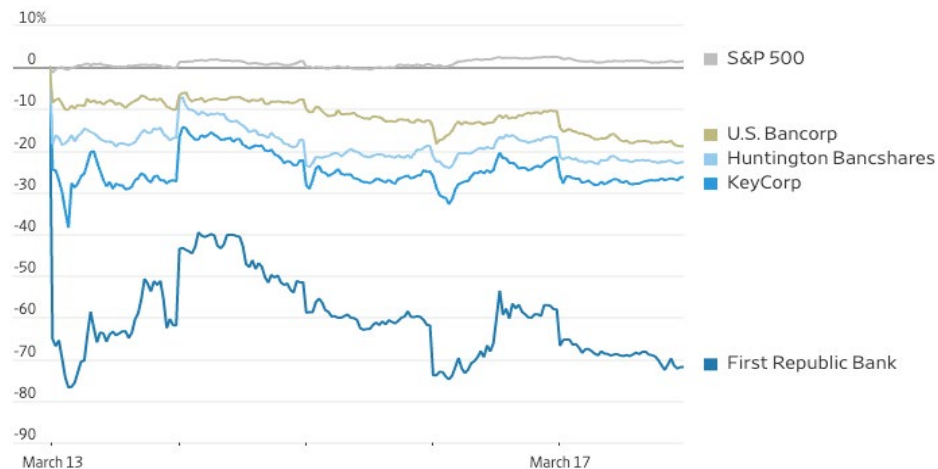
Relate to “News”

UBS Agrees to Buy Credit Suisse for More Than \$3 Billion

Deal is part of effort to prevent further erosion of confidence in banking system

Mar 19, 2023 – Wall Street Journal

[UBS Group](#) AG [UBS -5.50%](#) ▼ agreed to take over its longtime rival [Credit Suisse Group](#) AG [CS -6.94%](#) ▼ for more than \$3 billion, [pushed into the biggest banking deal](#) in years by regulators eager to halt a dangerous decline in confidence in the global banking system.



Source: FactSet

Mar 18, 2023 – Wall Street Journal

What you should be able to do?

Relate to “News”

Stocks Fall as Tariff Fears Ripple Through Economy

Major indexes had found their footing by midafternoon, then resumed their decline

By [Karen Langley](#) [Follow](#) and [Katy Barnato](#)

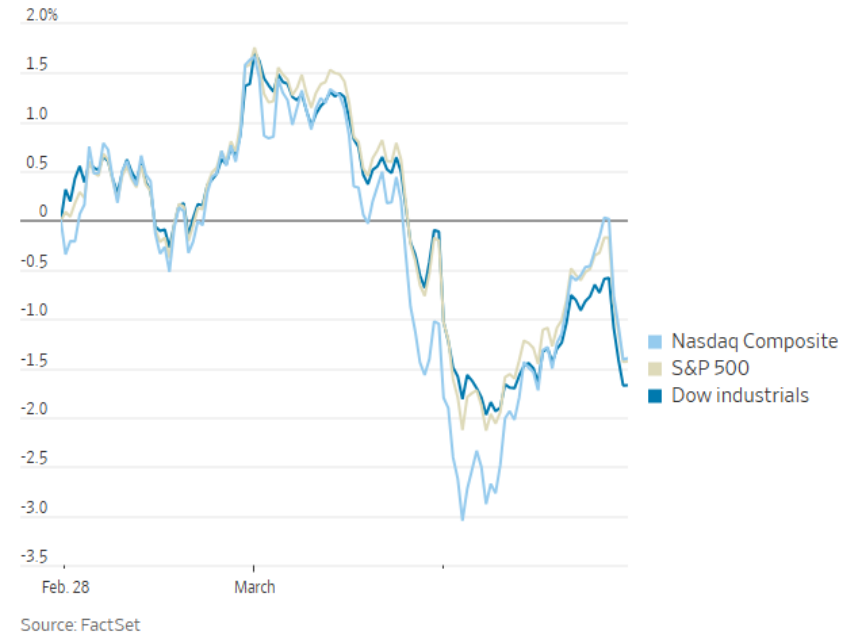
Mar 4, 2024 – Wall Street Journal

Briefing | When the tide turns

Rising interest rates and inflation have upended investing



Index performance, past three sessions



The breadth of losses is even more striking than the depth. In particular, the “60/40 portfolio”, comprising 60% stocks and 40% bonds, a popular choice for investors seeking a good return without running big risks, has performed appallingly. Evan Brown and Louis Finney of the asset-management arm of UBS, a Swiss bank, calculate that, as of mid-October, a 60/40 portfolio of American equities and Treasuries had had its worst year since 1937.

Dec 8, 2022 – The Economist

What you should be able to do?

Relate to “News”

Betting Against Banks Brings Reward and Backlash

Rising interest rates have made short selling fashionable again

By [Matt Wirz](#) [Follow](#) and [Ben Foldy](#) [Follow](#)

May 9, 2023 9:00 am ET

Short selling is back in vogue on Wall Street, rewarding some who bet against banks while triggering a backlash.

Short selling, or betting that securities prices will fall, surged over the past year as many investors braced for market turbulence. The approach helped hedge funds beat major stock indexes last year for the first time since 2008.

Controversy erupted in recent weeks, with critics alleging that traders are spooking bank customers into withdrawing deposits all at once. Lawyers representing the banking industry called last week for regulators to ban short sales of financial-company shares.

Mar 9, 2023 – Wall Street Journal

What is “shorting” a stock?

SPAC Mania Gives Early Investors Steady Returns With Little Risk

Hedge funds score gains when blank-check companies rise after announcing deals to take startups public

By [Amrith Ramkumar](#)

Jan. 13, 2021 5:30 am ET

Jan 13, 2021 – Wall Street Journal

SPAC Mania Is Dead. The SEC Wants to Keep It That Way.

New rules aim to make special-purpose acquisition company deals more like traditional IPOs

By [Paul Kiernan](#) [Follow](#)

Updated Jan. 24, 2024 12:41 pm ET

WASHINGTON—The [speculative bonanza](#) in special-purpose acquisition companies, better known as SPACs, appears to be dead. Gary Gensler wants to make sure it doesn't come back to life.

Jan 24, 2024 – Wall Street Journal

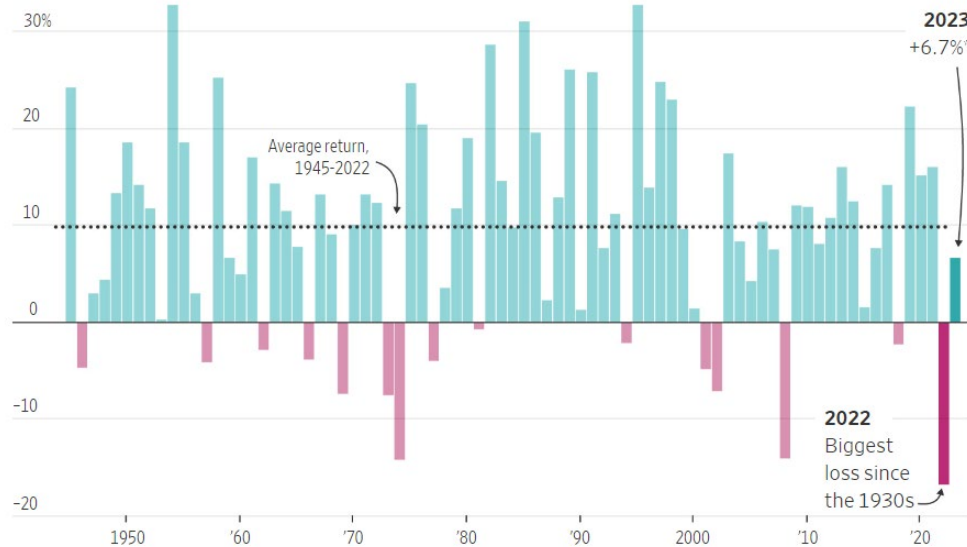
What you should be able to do?

Relate to “News”

The Trusted 60-40 Investing Strategy Just Had Its Worst Year in Generations

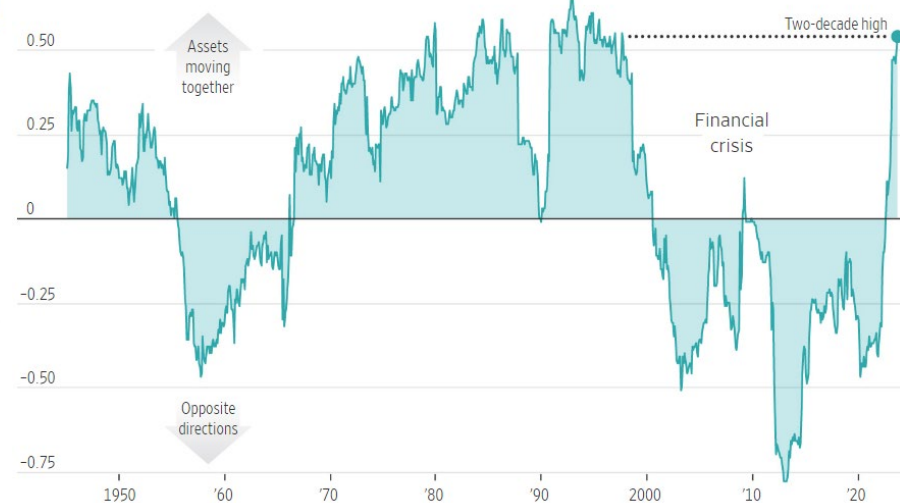
Higher interest rates and inflation are upending millions of Americans' retirement planning. Wall Street's boilerplate mix of stocks and bonds isn't cutting it anymore.

Annual return for a portfolio in 60% stocks and 40% bonds



*Figure for 2023 is through September.
Source: Leuthold Group

Correlation of returns for the S&P 500 and long-term Treasury bonds



Note: Three-year rolling correlation, monthly price changes.
Source: Standpoint Asset Management

FINANCE | STOCKS

Stocks Drop After Hot Inflation Report Clouds Rate-Cut Outlook

Tuesday's session was the worst CPI day for stocks since 2022

By Eric Wallerstein

Oct. 19, 2023 5:30 am ET

Oct 19, 2023 – Wall Street Journal

By Charley Grant [Follow](#)
Feb. 13, 2024 4:41 pm ET

Feb 13, 2024 – Wall Street Journal

What you should be able to do?

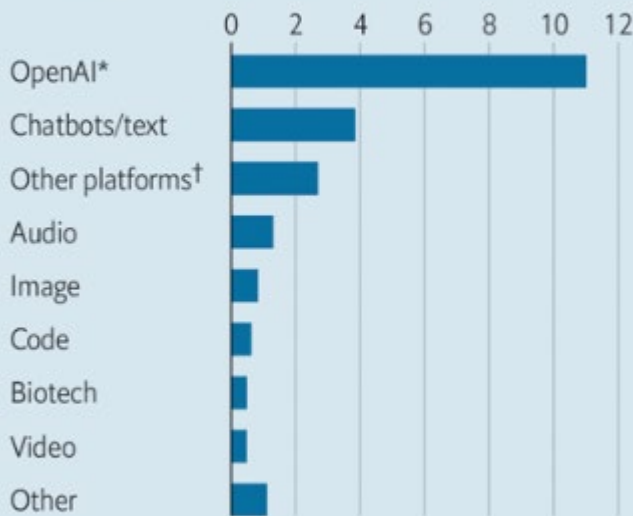
Relate to “News”

Investors are going nuts for ChatGPT-ish artificial intelligence

Even Elon Musk wants his own AI chatbot

Generation gain

Generative-AI startups, cumulative capital raised
By category, February 2023, \$bn



*Includes reported \$10bn from Microsoft †Systems like OpenAI on top of which other services can be built

Sources: NfX; press reports

The Economist

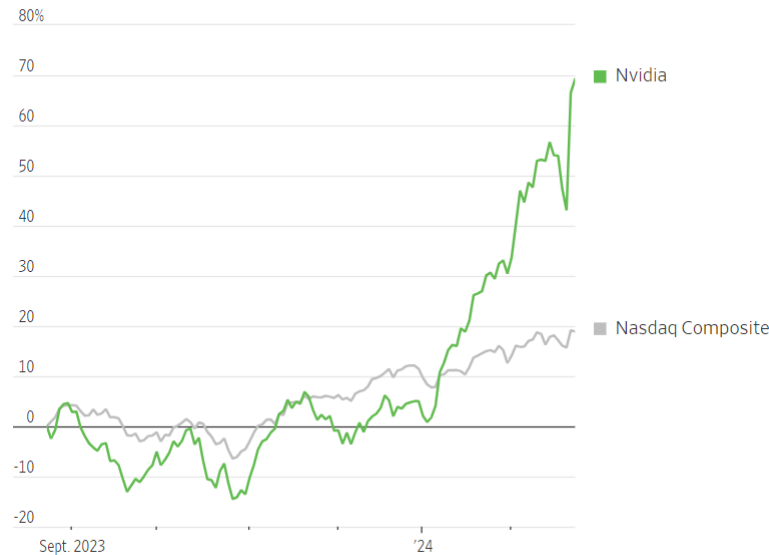
Feb 28, 2023 – The Economist

Surging Nvidia Stock Keeps Drawing In More Believers

The chip company that is cashing in on the market's artificial-intelligence obsession seems to many investors like an unstoppable force

Feb 24, 2024 – Wall Street Journal

Index and stock performance



Source: FactSet

TECHNOLOGY | ARTIFICIAL INTELLIGENCE | STREETWISE

How to Lose Money on the World's Most Popular Investment Theme

Pity the investors in the three artificial-intelligence-themed ETFs that managed to lose money this year

By [James Mackintosh](#) [Follow](#)

Sep 1, 2024 – The Wall Street Journal

What you should be able to do?

Relate to “News”

Melvin Capital, GameStop and the road to disaster

Hedge fund faces questions over risk management after losing half its money in a Reddit trading frenzy

Feb 6, 2021 – Financial Times

GameStop shares did indeed go 'to the moon'. And back.

Share price at market close (\$)

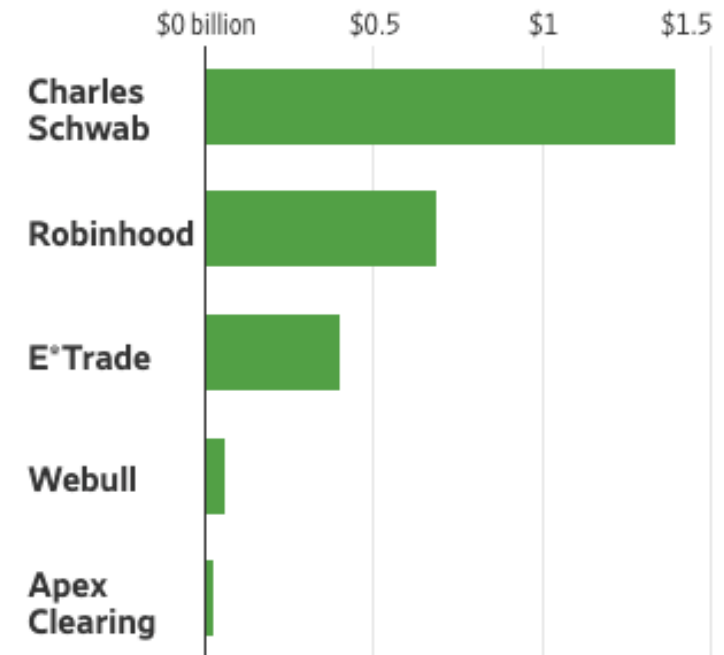


GameStop Mania Drives Scrutiny of Payments to Online Brokers

Payment for order flow enabled zero-commission trades but has long been controversial

Feb 4, 2021 – Wall Street Journal

Payment-for-order-flow revenue in 2020, by brokerage



What you should be able to do?

Relate to “News”

Why A Data Revolution Is Giving Socially Responsible Investors an Edge

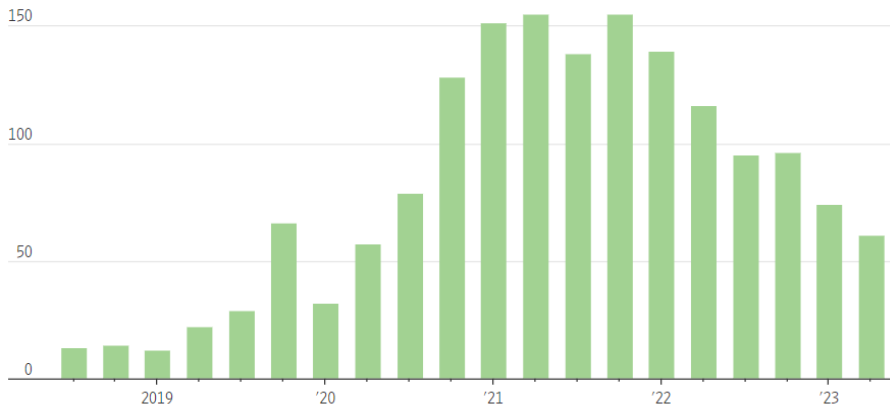
By [Matthew Heimer](#) September 25, 2019

It's getting a lot easier to align your financial strategy with your conscience. A report issued to clients this week by BofA Merrill Lynch Global Research makes an unusually strong data-driven case for “ESG investing”—taking companies’ environmental impact, social impact, and governance into account when deciding whether to buy their stocks. The report’s takeaway: If you’re not using ESG analysis, you may be leaving money on the table.

The Latest Dirty Word in Corporate America: ESG

Executives switch to alternatives like ‘responsible business’ to describe corporate initiatives

Number of S&P 500 companies citing ‘ESG’ on earnings calls



Source: FactSet

Climate change will crush real estate values for investors who don't prepare, new report says

PUBLISHED MON, APR 8 2019·12:11 PM EDT | UPDATED FRI, MAY 17 2019·4:19 PM EDT



Diana Olick
[@IN/DIANAOLICK](#)
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- Big real estate firms are pouring resources into calculating climate risk and its likely effect on property portfolios — everything from increasingly extreme weather to sea-level rise.
- “This process will be painful for investors who are caught off guard, but those who are prepared have the potential to outperform,” a new report from the Urban Land Institute says.

Jan 9, 2024 – Wall Street Journal

By [Chip Cutter](#) [Follow](#) and [Emily Glazer](#) [Follow](#)

Updated Jan. 9, 2024 5:20 pm ET

Goals of the Course

- A “Tool-kit”
 - To evaluate investment decisions
 - To analyze financing decisions
 - To apply in more complex situations
- Fundamental decision making skills
 - Banking
 - Consulting
 - Marketing
 - Policy work